

```

1  ///! structures :
2
3  /*
4      ? what is structure (struct) in c++ :
5      ~ structure is a user defined data type
6      % way to group several related data / variables together/ into one place
7      ! it is a collection of variables of different types under a single name
8      * each variable in a structure is called a member of the structure
9      that can hold multiple values of different types
10     - it is used to group related data together
11     - it is similar to class but it does not have member functions and access specifiers
12 */
13
14 ///! the syntax of structure :
15 /*
16     struct structure_name {
17         data_type member1;
18         data_type member2;
19         ...
20     };
21 */
22
23 //ex :
24 struct Person {
25     string name;
26     int age;
27     double height;
28 };
29

```

```

30 ////////////////////////////////////
31 ///! to access the members of the structure we can use the dot operator (.) :
32
33 int main() {
34     // creating a structure variable
35     Person person1;
36
37     // assigning values to the members of the structure
38     person1.name = "John";
39     person1.age = 30;
40     person1.height = 1.75;
41
42     // accessing the members of the structure
43     cout << "Name: " << person1.name << endl;
44     cout << "Age: " << person1.age << endl;
45     cout << "Height: " << person1.height << endl;
46
47     return 0;
48 }
49 ////////////////////////////////////
50

```

```

51  //!< you cn crete multiple structure variables of the same type or different types :
52
53  //!< the syntax of structure :
54  /*
55      struct structure_name_1 {
56          data_type member1;
57          data_type member2;
58          ...
59      };
60
61      struct structure_name {
62          data_type member1;
63          data_type member2;
64          // structure_name_1 member3; // structure as a member of another structure
65          ...
66      };
67  */
68

```

```

69  //ex : 🐼
70  struct Address {
71      string city;
72      string country;
73  };
74  struct person
75  {
76      string name;
77      int age;
78      double height;
79      Address address; // structure as a member of another structure
80  };
81
82

```

```
83  int main()
84  {
85      // creating a structure variable
86      person person1;
87
88      // assigning values to the members of the structure
89      person1.name = "John";
90      person1.age = 30;
91      person1.height = 1.75;
92      person1.address.city = "New York";
93      person1.address.country = "USA";
94
95      // accessing the members of the structure
96      cout << "Name: " << person1.name << endl;
97      cout << "Age: " << person1.age << endl;
98      cout << "Height: " << person1.height << endl;
99      cout << "City: " << person1.address.city << endl;
100     cout << "Country: " << person1.address.country << endl;
101
102     return 0;
103 }
    ——— FUNCTION LINES : 18 ———
104
105
```